

TAICHUNG WUQI, Taiwan

WMO: 467770

Lat: 24.2561N

Lon: 120.5233E

Elev: 32

StdP: 100.94

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.3	(c) 11.4	(d) 4.4	(e) 5.2	(f) 11.8	(g) 5.8	(h) 5.7	(i) 12.7	(j) 16.2	(k) 14.1	(l) 15.2	(m) 15.2	(n) 7.8	(o) 30	(p) 0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.0	(c) 32.9	(d) 27.3	(e) 32.3	(f) 27.2	(g) 31.8	(h) 27.0	(i) 28.0	(j) 31.9	(k) 27.7	(l) 31.5	(m) 27.3	(n) 31.2	(o) 6.0	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	22.7	31.1	26.5	22.1	30.9	26.2	21.6	30.7	90.1	31.9	88.5	31.7	87.0	31.4	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.7	11.5	8.7	35.3	1.0	2.2	8.0	36.9	7.4	38.3	6.8	39.5	6.1	41.2		
			DB	6.1	28.7	1.3	0.4	5.1	29.0	4.3	29.3	3.5	29.5	2.6	29.8	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	16.4	16.3	18.8	22.5	25.5	28.0	29.2	29.0	27.4	24.6	21.9	18.9
	DBStd	5.07	2.18	2.54	3.07	3.33	2.36	1.96	1.16	1.12	1.73	1.80	2.09	2.35
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	193	65	66	32	7	1	0	0	0	0	0	2	20
	CDD10.0	4836	200	177	273	374	480	538	596	589	522	453	357	276
	CDD18.3	1986	7	10	46	131	222	288	338	331	272	194	109	38
	CDH23.3	19212	6	21	146	795	1911	3327	4351	4172	2904	1202	346	31
(13)	CDH26.7	6683	0	1	14	117	461	1244	1940	1754	946	185	18	2
(14)	Wind	WSAvg	5.2	6.9	6.4	5.5	4.6	3.8	4.0	3.7	4.6	6.6	6.1	6.9
(15) Precipitation	PrecAvg	1506	34	84	97	144	195	268	218	238	144	31	25	30
	PrecMax	2246	107	337	206	517	481	669	582	594	721	232	167	84
	PrecMin	696	2	3	30	6	23	23	35	25	12	1	0	1
	PrecStd	419	23	81	53	114	124	160	149	163	186	51	38	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.3	25.8	28.3	30.3	31.8	33.4	33.7	33.4	32.4	30.2	28.1	25.7
		MCWB	19.7	21.1	22.6	24.9	26.3	27.4	27.4	27.4	26.9	25.7	23.7	21.5
	2%	DB	22.0	23.6	26.4	29.1	30.8	32.5	32.9	32.7	31.6	29.2	27.1	24.0
		MCWB	18.5	19.7	22.0	24.6	26.1	27.1	27.4	27.4	26.6	25.0	23.0	20.9
	5%	DB	20.7	21.9	24.9	28.1	30.0	31.7	32.3	32.2	31.0	28.5	26.2	22.9
		MCWB	17.6	18.9	21.1	23.9	25.7	27.0	27.3	27.2	26.3	24.2	22.2	19.8
	10%	DB	19.6	20.3	23.6	27.1	29.1	31.1	31.8	31.6	30.4	27.7	25.3	22.1
		MCWB	16.8	17.4	20.4	23.4	25.1	26.8	27.1	27.0	25.9	23.5	21.6	19.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	21.7	23.4	25.6	27.2	28.3	28.4	28.5	27.6	26.4	24.5	22.3
		MCDB	23.4	24.7	27.3	29.3	30.7	32.2	32.2	32.2	31.4	29.5	27.3	24.9
	2%	WB	18.8	20.4	22.3	24.9	26.5	27.7	28.0	28.0	27.1	25.5	23.6	21.2
		MCDB	21.5	23.3	25.9	28.5	30.1	31.6	31.8	31.9	30.9	28.5	26.5	23.4
	5%	WB	18.0	18.9	21.5	24.3	26.0	27.3	27.7	27.6	26.6	24.8	22.9	20.2
		MCDB	20.5	21.5	24.7	27.7	29.5	31.2	31.6	31.5	30.4	27.9	25.8	22.3
	10%	WB	17.1	17.7	20.5	23.6	25.4	26.8	27.4	27.2	26.2	24.1	22.1	19.5
		MCDB	19.4	20.1	23.4	26.9	28.7	30.7	31.3	31.1	29.9	27.3	24.8	21.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.0	5.4	5.2	5.0	4.8	5.0	4.9	5.4	5.4	5.7	5.3
		MCDBR	6.7	7.7	7.7	5.8	5.5	5.5	5.5	5.5	5.6	5.8	6.6	6.2
		MCWBR	4.7	5.2	4.5	3.7	3.8	3.5	3.3	3.4	3.4	3.9	4.4	4.3
	5% WB	MCDBR	6.5	6.8	6.5	5.5	5.3	5.2	5.1	5.1	5.3	5.1	6.0	5.8
		MCWBR	4.7	5.0	4.4	3.9	3.8	3.6	3.2	3.4	3.4	3.7	4.3	4.1
(40) Clear-Sky Solar Irradiance	taub		0.508	0.566	0.684	0.692	0.588	0.537	0.550	0.583	0.576	0.542	0.514	0.508
	taud		1.970	1.853	1.628	1.663	1.937	2.095	2.050	1.950	1.943	1.981	2.036	1.987
	Ebn at Noon		735	722	660	669	741	773	763	739	733	734	726	715
	Edh at Noon		164	197	258	254	192	163	170	188	185	171	153	156
(44) All-Sky Solar Radiation	RadAvg		2.92	3.21	3.73	4.12	4.75	5.21	5.93	5.37	4.95	4.38	3.27	2.90
	RadStd		0.41	0.47	0.41	0.55	0.71	0.56	0.36	0.52	0.47	0.36	0.31	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	+0.25	N/A	N/A	+0.22	N/A	N/A	N/A	N/A	+127	+112

Nomenclature: See separate page